

Answer Key

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Exam Duration: 120 minutes

Total Marks: 100 points

Section A: Multiple Choice Questions (2 points each, 30 points total)

1. Which phenomenon aligns with the core idea of Moravec's Paradox? ()

A. Computers solve complex mathematical equations quickly but struggle to recognize ambiguous handwritten text

B. Humans easily perform logical reasoning, while computers require complex algorithms

C. Computers precisely control robot movement, but humans often make movement errors

D. Humans struggle to memorize large amounts of data, while computers store it efficiently
2. Which description of Nash Equilibrium is correct? ()

A. At Nash Equilibrium, the total payoff of all participants is maximized

B. At Nash Equilibrium, unilaterally changing strategies reduces one's own payoff

C. Zero-sum games have no Nash Equilibrium

D. Nash Equilibrium is always a globally optimal solution
3. Given $P(B)=0.6$ and $P(A|B)=0.5$, the joint probability $P(A,B)$ is ()

A. 0.3 B. 0.6 C. 0.8 D. 1.1
4. According to Zadeh's fuzzy logic operators, if $A=0.7$ and $B=0.4$, the result of $A \text{ OR } B$ is ()

A. 0.3 B. 0.4 C. 0.7 D. 1.1
5. The common mutation method for binary encoding in genetic algorithms is ()

A. Value replacement B. Bit flip C. Sequence rearrangement D. Segment exchange
6. The main purpose of using ring topology in Particle Swarm Optimization (PSO) is to ()

A. Accelerate convergence speed B. Increase population diversity C. Simplify computational complexity D. Improve global search accuracy

7. The core function of an activation function in neural networks is to ()

A. Reduce model complexity B. Introduce non-linearity C. Accelerate gradient descent D. Reduce overfitting

8. The core advantage of convolutional layers for image feature extraction is ()

A. Fully connecting all inputs B. Sharing weights to reduce parameters C. Directly outputting classification results D. Alleviating vanishing gradients

9. Residual connections mainly solve which problem of deep networks through "skip connections"? ()

A. Overfitting B. Vanishing gradients C. Excessive computational load D. Parameter redundancy

10. The core advantage of LSTM over vanilla RNN is ()

A. Faster computation speed B. Fewer model parameters C. Better handling of long-sequence dependencies D. Applicability to image data

11. When the query sequence length is n and embedding depth is d , the output dimension of cross-attention is ()

A. $n \times n$ B. $n \times d$ C. $d \times d$ D. $n \times 1$

12. The core setup of unsupervised domain adaptation is ()

A. Both source and target domains have labels B. Source domain has labels, target domain has no labels

C. Source domain has no labels, target domain has labels D. Neither source nor target domain has labels

13. The core idea of diffusion models is ()

A. Adversarial training between generator and discriminator B. Training a noise predictor and denoising incrementally

C. Direct sampling generation based on probability distributions D. Capturing global dependencies using self-attention

14. In Hidden Markov Models (HMM), the element a_{ij} of the state transition matrix A represents ()
- A. Transition probability from observation state i to observation state j B. Transition probability from hidden state i to hidden state j
- C. Probability of hidden state i corresponding to observation state j D. Probability of observation state i corresponding to hidden state j
15. In Q-learning, the "exploration rate" ϵ in the epsilon-greedy strategy represents ()
- A. Probability of selecting the optimal action B. Probability of randomly selecting an action
- C. Step size of Q-value update D. Discount factor for future rewards

Section B: Fill-in-the-Blanks (1 point each, 10 points total)

1. The core feature of a zero-sum game is ___, meaning an increase in one party's payoff is necessarily accompanied by a decrease in the other party's payoff.
2. In fuzzy logic, the calculation formula for NOT A is ___, reflecting the negation of a fuzzy set.
3. In neural network regularization, L2 regularization prevents overfitting by ___, while Dropout achieves this by ___.
4. The formula for calculating the number of learnable parameters in a CNN convolutional layer is: $(\text{kernel_size} \times \text{in_channels} + \text{bias}) \times \text{___}$.
5. The core training objective of CLIP is to maximize the similarity between ___ to achieve cross-modal alignment.
6. The core of chain-of-thought prompting is to guide large language models to ___ to solve complex problems.
7. The stationary distribution π of a Hidden Markov Model satisfies ___, meaning the distribution remains unchanged once stabilized.
8. The structure of a Q-table is ___, storing the value estimate for each state-action pair.
9. In reinforcement learning, the closer the discount factor γ is to 1, the more the agent ___ (values/ignores) future rewards.

Section C: Short Answer Questions (5 points each, 20 points total)

1. Briefly describe three common encoding methods in genetic algorithms and their characteristics.
2. Explain the core working principle of the self-attention block in Transformers.
3. Describe the loss function objective and role of the discriminator in Generative Adversarial Networks (GANs).

- Interpret the core update formula of Q-learning and explain the meaning of each parameter.

Section D: Calculation Questions (10 points)

Given a Hidden Markov Model with state transition matrix ($A = \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$), initial state distribution ($\pi = [0.5, 0.5]$), and observation probability matrix ($B = \begin{bmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{bmatrix}$).

Calculate:

- The occurrence probability of the hidden state sequence "0→1"; (4 points)
- The stationary distribution π of the model (satisfying $\pi = \pi A$). (6 points)

Section E: Comprehensive Application Questions (30 points)

(I) Reinforcement Learning Q-learning Application (15 points)

Consider a 3×3 grid world with the agent's states as grid positions (9 states total). Actions include up, down, left, and right (4 actions). The target state is (3,3), with a +10 reward for reaching the target, a -1 reward for hitting walls (moving outside boundaries), and a -0.5 reward for other actions.

- Draw a schematic diagram of the Q-table structure and explain its function; (5 points)
- If the current $Q((2,2), \text{Right}) = 3.2$, executing the "Right" action reaches the target state, with learning rate $\alpha = 0.1$ and discount factor $\gamma = 0.9$, calculate the updated Q-value using the Q-learning update formula; (5 points)
- Explain how the epsilon-greedy strategy balances exploration and exploitation in this scenario. (5 points)

(II) Deep Learning Application (15 points)

An image classification task uses a CNN architecture with input as 3-channel 28×28 images. The network structure is as follows:

Conv1 (3×3 kernel, 16 output channels, padding=1) → BatchNorm → ReLU → MaxPool (2×2, stride=2) → Conv2 (3×3 kernel, 32 output channels, padding=1) → BatchNorm → ReLU → GlobalAvgPool → FC (10 output classes).

- Calculate the number of learnable parameters for Conv1 and Conv2 layers; (6 points)
 - Explain the function and working principle of the BatchNorm layer; (5 points)
 - If overfitting occurs during training, list two effective solutions and explain their principles. (4 points)
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Instructions for Word Formatting

1. Copy the entire content above into Microsoft Word.
 2. Use "Heading 1" for the title ("Computational Intelligence Exam Paper"), "Heading 2" for section titles (Section A-E), and "Normal" for body text.
 3. Adjust the formula layout (e.g., state transition matrix) using Word's Equation Editor for better readability.
 4. Add page numbers, margins, and font formatting (recommended: Calibri, 11pt) as needed for academic use.
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Section A: Multiple Choice Questions (2 points each, 30 points total)

1. A
2. B
3. A (Calculation: $P(A,B) = P(A|B) \times P(B) = 0.5 \times 0.6 = 0.3$)
4. C (Zadeh's OR = $\max(A,B) = \max(0.7,0.4) = 0.7$)
5. B
6. B
7. B
8. B
9. B
10. C
11. B
12. B
13. B
14. B
15. B

Section B: Fill-in-the-Blanks (1 point each, 10 points total)

1. total payoff is zero
2. 1 - A
3. penalizing large weights; randomly deactivating neurons
4. out_channels
5. relevant image and text embeddings
6. reason step by step

7. $(\pi = \pi_A)$
8. number of states $\times$ number of actions
9. values

Section C: Short Answer Questions (5 points each, 20 points total)

1. Three Encoding Methods in Genetic Algorithms:

- Binary Encoding: Simple and intuitive, uses 0s and 1s; mutation via bit flip, suitable for discrete problems.
- Value Encoding: Directly uses real numbers; flexible for continuous optimization, requires custom mutation rules.
- Permutation Encoding: Based on sequence ordering; mutation via segment exchange, suitable for ordering problems (e.g., TSP).

2. Core Working Principle of Self-Attention Block:

Convert input into Query (Q), Key (K), Value (V) matrices; compute scaled dot-product of Q and K to get attention weights; normalize weights with SoftMax; output via weighted sum of V, realizing global element correlation.

3. GAN Discriminator's Loss Function Objective & Role:

- Objective: Maximize the probability of correctly classifying real samples as "real" and generated samples as "fake".
- Role: Guide the generator to produce more realistic data by learning discriminative features between real and generated samples.

4. Q-learning Core Update Formula & Parameter Explanation:

Formula: $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$

- $Q(s,a)$: Current value estimate of action (a) in state (s).
- α : Learning rate, controls Q-value update step size.
- r : Immediate reward from taking action (a) in state (s).
- γ : Discount factor, weights the importance of future rewards.
- s' : Next state after taking action (a).
- $\max_{a'} Q(s',a')$: Maximum Q-value of all possible actions in next state (s').

Section D: Calculation Questions (10 points)

1. Probability of Hidden State Sequence "0→1":

($P = \pi(0) \times A(0,1) = 0.5 \times 0.4 = 0.2$) (4 points)

2. Stationary Distribution (π):

Let ($\pi = [\pi_0, \pi_1]$), satisfy ($\pi = \pi A$) and ($\pi_0 + \pi_1 = 1$):

- ($\pi_0 = 0.6\pi_0 + 0.3\pi_1$)
- ($\pi_1 = 0.4\pi_0 + 0.7\pi_1$)
- Solve: ($\pi_0 = \frac{3}{7} \approx 0.4286$), ($\pi_1 = \frac{4}{7} \approx 0.5714$) (6 points)

Section E: Comprehensive Application Questions (30 points)

(I) Reinforcement Learning Q-learning Application (15 points)

1. Q-table Structure & Function:

- Structure: 9 rows (3×3 grid states) $\times$ 4 columns (up/down/left/right actions), each cell stores ($Q(s,a)$).
- Function: Records the long-term cumulative reward estimate of each state-action pair, providing a basis for action selection. (5 points)

2. Updated Q-value Calculation:

Target Q-value = ($r + \gamma \max_{a'} Q(s',a') = 10 + 0.9 \times 0 = 10$) (target state has no future rewards)

Updated ($Q = 3.2 + 0.1 \times (10 - 3.2) = 3.88$) (5 points)

3. Epsilon-greedy Strategy for Exploration-Exploitation Balance:

- High (ϵ) (e.g., 0.2): 20% random exploration (discover new actions) + 80% exploitation (select current optimal action).
- Reduce (ϵ) over time: Decrease exploration, focus on exploiting learned optimal strategies. (5 points)

(II) Deep Learning Application (15 points)

1. Learnable Parameters of Conv1 & Conv2:

- Conv1: ($(3 \times 3 \times 3 + 1) \times 16 = (27 + 1) \times 16 = 448$)
- Conv2: ($(3 \times 3 \times 16 + 1) \times 32 = (144 + 1) \times 32 = 4640$) (6 points)

2. BatchNorm Layer: Function & Working Principle:

- Function: Stabilize training, accelerate convergence, alleviate internal covariate shift.

- Principle: Normalize each batch of data to mean=0, variance=1; adjust distribution via learnable parameters (γ) (scale) and (β) (shift). (5 points)

3. Solutions for Overfitting & Principles:

- Dropout: Randomly deactivate neurons during training, reduce interdependence between neurons, enhance model generalization.
 - L2 Regularization: Penalize large weights, reduce model complexity, avoid overfitting to noise in training data. (4 points)
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Note:

- For short answer questions, partial scores are given for key points.
- For calculation questions, scores are deducted for incorrect formulas or miscalculations.

| (注：文档部分内容可能由 AI 生成)